



NCEA Level 3 Calculus

AS 91578
Differentiation

Learning Workbook
2026 Edition

Achievement Standard 91578

Apply Differentiation Methods in Solving Problems

Achievement Criteria

Achievement	Achievement with Merit	Achievement with Excellence
Apply differentiation methods in solving problems.	Apply differentiation methods, using relational thinking, in solving problems.	Apply differentiation methods, using extended abstract thinking, in solving problems.

Notes

1. Apply differentiation methods in solving problems involves:

- selecting and using methods
- demonstrating knowledge of concepts and terms
- communicating using appropriate representations.

Relational thinking involves one or more of:

- selecting and carrying out a logical sequence of steps
- connecting different concepts or representations
- demonstrating understanding of concepts
- forming and using a model;

and also relating findings to a context, or communicating thinking using appropriate mathematical statements.

Extended abstract thinking involves one or more of:

- devising a strategy to investigate or solve a problem
- identifying relevant concepts in context
- developing a chain of logical reasoning, or proof
- forming a generalisation;

and also using correct mathematical statements, or communicating mathematical insight.

2. Problems are situations that provide opportunities to apply knowledge or understanding of mathematical concepts and methods. Situations will be set in real-life or mathematical contexts.
3. Methods are selected from those related to:
 - derivatives of power, exponential, and logarithmic (base e only) functions
 - derivatives of trigonometric (including reciprocal) functions
 - optimisation
 - equations of normals
 - maxima and minima and points of inflection
 - related rates of change
 - derivatives of parametric functions
 - chain, product, and quotient rules
 - properties of graphs (limits, differentiability, continuity, concavity).

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Chapter 1

Limits, Continuity and Differentiability

LEARNING OBJECTIVES

By the end of this chapter, you should be able to:

1. **Evaluate Limits:** Distinguish between the “journey” (limit) and the “destination” (function value) using algebraic and graphical methods.
2. **Graphical Analysis:** Read and interpret limits from complex functions, identifying holes, jumps, and asymptotes.
3. **Analyse Continuity:** Apply the formal mathematical “Three-Step Test” to verify if a function is continuous.
4. **First Principles:** Derive gradient functions from the definition of a limit to understand the origin of differentiation.
5. **Differentiability:** Explain the relationship between continuity and differentiability, identifying where gradient functions fail.

TOPIC 1 · THE CONCEPT OF A LIMIT

In Calculus, a limit describes the behaviour of a function $f(x)$ as the independent variable x approaches a specific constant a , without necessarily reaching that value. It focuses on the **limiting value** (the height) that the function is trending towards.

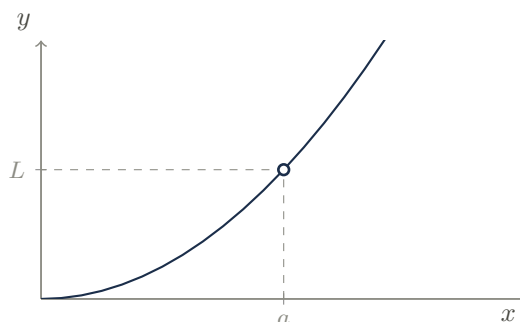
$$\lim_{x \rightarrow a} f(x) = L$$

KEY IDEA

The Prediction Strategy: When evaluating $\lim_{x \rightarrow a}$, do not be distracted by the specific point $x = a$. The limit is a prediction of the height based on the trajectory of the curve from both sides. It is valid even if there is a hole at the destination.

The Indeterminate Form (0/0)

Direct substitution often leads to $0/0$. This indicates that the limit exists but is hidden by a common factor representing a point-discontinuity (a hole). To resolve this, use “Algebraic Surgery”: Factorise, Rationalise, or Simplify.



EXAM TIP

When evaluating limits at infinity ($x \rightarrow \infty$), divide every term by the highest power of x in the denominator. Recall that any term $\frac{k}{x^n} \rightarrow 0$ as x grows large.

Example 1

Evaluate $\lim_{x \rightarrow 3} (x^2 - 2x)$.

Solution:

1. Since there are no fractions or square roots causing issues, we use direct substitution.
2. Substitute $x = 3$ into the expression:

$$3^2 - 2(3) = 9 - 6 = 3$$

3. The limit is 3.

Example 2

Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

Solution:

1. Try direct substitution: $\frac{5^2 - 25}{5 - 5} = \frac{0}{0}$. This indicates a hole.
2. Factorise the numerator (Difference of Two Squares):

$$\lim_{x \rightarrow 5} \frac{(x - 5)(x + 5)}{x - 5}$$

3. Cancel the common term $(x - 5)$, which removes the hole:

$$\lim_{x \rightarrow 5} (x + 5)$$

4. Now substitute $x = 5$:

$$5 + 5 = 10$$

Example 3

Evaluate $\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2}$.

Solution:

1. Substitution gives $\frac{0}{0}$. We must rationalise the denominator.
2. Multiply top and bottom by the conjugate $(\sqrt{x} + 2)$:

$$\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2} \times \frac{\sqrt{x} + 2}{\sqrt{x} + 2}$$

3. Simplify the denominator (Difference of Squares):

$$\lim_{x \rightarrow 4} \frac{(x - 4)(\sqrt{x} + 2)}{(\sqrt{x})^2 - 2^2} = \lim_{x \rightarrow 4} \frac{(x - 4)(\sqrt{x} + 2)}{x - 4}$$

4. Cancel $(x - 4)$ and evaluate:

$$\lim_{x \rightarrow 4} (\sqrt{x} + 2) = \sqrt{4} + 2 = 4$$

Example 4

Evaluate $\lim_{x \rightarrow 0} \frac{\frac{1}{x+2} - \frac{1}{2}}{x}$.

Solution:

1. Direct substitution gives $\frac{0}{0}$. We need to simplify the complex fraction.
2. Combine the fractions in the numerator using a common denominator of $2(x+2)$:

$$\frac{\frac{2-(x+2)}{2(x+2)}}{x} = \frac{2-x-2}{2x(x+2)} = \frac{-x}{2x(x+2)}$$

3. Cancel the x terms:

$$\lim_{x \rightarrow 0} \frac{-1}{2(x+2)}$$

4. Substitute $x = 0$:

$$\frac{-1}{2(0+2)} = -\frac{1}{4}$$

Example 5

Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 - 3x + 5}{2x^2 + 1}$.

Solution:

1. **Analyse:** As $x \rightarrow \infty$, the terms with the highest powers dominate. The $-3x$, $+5$, and $+1$ become insignificant.
2. **Divide by highest power:** Divide every term by x^2 .

$$\lim_{x \rightarrow \infty} \frac{\frac{4x^2}{x^2} - \frac{3x}{x^2} + \frac{5}{x^2}}{\frac{2x^2}{x^2} + \frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{4 - \frac{3}{x} + \frac{5}{x^2}}{2 + \frac{1}{x^2}}$$

3. **Evaluate Limit:** As $x \rightarrow \infty$, any term with x in the denominator approaches 0 (e.g., $\frac{3}{x} \rightarrow 0$).

$$\frac{4 - 0 + 0}{2 + 0} = \frac{4}{2} = 2$$

4. **Answer:** The limit is 2. (This means the graph has a horizontal asymptote at $y = 2$).